Practical Equivalent-Key Recovery in GRAFHEN

Remi Geraud-Stewart

Amazon

Abstract. GRAFHEN is a group-based homomorphic-encryption proposal whose public key is a rewriting system and whose secret key is a permutation representation. We present *Maverick*, an equivalent-key recovery attack. Maverick breaks every released GRAFHEN challenge, including the recommended two-copy S_{11} instance: it reconstructs an equivalent key from public rules and correctly decrypts all 20,000 supplied labelled ciphertexts. On 12 independently generated recommended-parameter keys, the median end-to-end time is 552 s and the median peak memory use is 11.28 GB on an Apple M3 Pro.

The attack converts selected public rewrite rules into group relators, reconstructs a regular action by Todd–Coxeter enumeration, recognizes the resulting permutation representations, and aligns the two sides through the public mixed relations. Public labelled encryptions then calibrate an equivalent decryptor. We prove a proof-carrying version of this procedure: a target-order table with a replayable trace certifies the recovered regular action. The reported S_{11} experiments use target-order closure checks, complete-corpus verification, and decryptor validation. We also give an output-sensitive analysis and state the hypotheses needed to extrapolate beyond the measured instances.

Following an independent key-recovery attack, GRAFHEN proposed in July 2026 to replace S_{11} by $\mathrm{PSL}_2(343)$. We adapt Maverick to this setting and demonstrate complete public-rule recovery, alignment, certification, and calibration on generated $\mathrm{PSL}_2(169)$ instances. The two-side $q = 169$ run decrypts all 5,000 held-out ciphertexts in a 103 s critical path using 0.93 GiB peak memory; a $k = 2$ admissibility-filtered corpus also succeeds. Under GRAFHEN's stated worst-case estimate for Dumezy's degree-based search, this degree-170, $d = 5$ setting already has cost $O(2^{850})$, well beyond the intended reach of that attack. At the target group, the post-enumeration recovery path from a generated regular action takes 35.0 s and 1.92 GiB peak memory. These results suggest that the proposed platform-group change is insufficient to rule out Maverick; recovery from an admissibility-filtered corpus at $q = 343$ remains to be measured because the pre-filter KeyGen enumeration exceeded 32 GiB of memory before it could produce a corpus.

1 Introduction

GRAFHEN [8] is a group-based homomorphic-encryption proposal in which ciphertexts are words and the public key is a non-confluent rewriting system. Evaluation is exact group composition, so the construction has no analogue of

the noise growth that drives bootstrapping in lattice-based FHE. The original recommended instantiation used two copies of S_{11}, each represented by five secret generators. Its security discussion estimated approximately $11!^4 > 2^{101}$ inequivalent keys and treated recovery of the hidden permutation representation as infeasible.

We show that selected public relations expose an equivalent representation. Our attack, *Maverick*, treats the per-side rules as group relators and reconstructs a regular action of each copy of S_n. Constructive recognition then recovers natural degree-n actions. The public mixed relations align the two independently recovered coordinate systems, yielding a simultaneous conjugate of the secret tuple. Public labelled encryptions calibrate the decryption kernel in that recovered frame, and hence an equivalent decryptor. This differs from the Todd–Coxeter attack model considered in [8, Section 6.2]: Maverick does not guess a finite-index subgroup of the combined presentation. It enumerates the two per-side presentations at the known regular degree $|S_n|$ and uses the mixed relations only for coordinate alignment.

Experimental result. At the original recommended parameters $(n, d) = (11, 5)$, Maverick succeeds on all 12 independently generated keys in our test set. Median wall time is 552 s and median peak resident memory is 11.28 GB on an Apple M3 Pro. We also evaluate Maverick on the independently generated public GRAFHEN challenge. It recovers every released key, including the recommended two-copy S_{11} instance, and the recovered decryptor classifies all 20,000 labelled ciphertexts in each instance. These end-to-end measurements combine exact relation checks with successful decryption.

Formal statement and experimental evidence. Theorem 1 gives the proof-carrying recovery theorem: a target-order Todd–Coxeter table accompanied by a replayable trace certifies the regular action, after which the remaining stages are publicly checkable. Proposition 2 proves the calibration step under its stated sampling condition. The reported S_{11} experiments use target-order closure checks, complete-corpus verification, and held-out labelled ciphertexts; they do not emit replay traces. Their resource use is output sensitive in the number and length of selected relators and in the peak coset-table size. The later scaling discussion explicitly identifies the additional hypotheses needed to extrapolate beyond the measured instances.

Contributions. The paper makes the following claims.

- A practical break of the original S_{11} construction: selected public-rule subsets recover the key, the complete corpus verifies it, and the recovered decryptor succeeds on independently generated keys and every released challenge.
- A centreless-group recovery theorem: target-order certificates and complete mixed conjugation relations uniquely align two independently reconstructed regular actions up to a common automorphism (Theorem 2).

– A group-independent calibration result that converts a common automorphic image into an equivalent decryptor from public labelled encryptions (Proposition 2).
– Evidence that the July 2026 platform-group revision does not yet rule out Maverick. A generated two-side $\mathrm{PSL}_2(169)$ control recovers and decrypts all 5,000 held-out ciphertexts without error in a 103 s critical path at 0.93 GiB peak memory. A $k = 2$ admissibility-filtered corpus at the same parameter also reaches the target order. The target-size downstream experiment completes in 35.0 s at 1.92 GiB; target-size filtered-corpus recovery remains unmeasured. Appendix E contains the group-specific construction, proofs, and full measurements.

Organisation. Section 2 fixes notation and describes the targeted GRAFHEN version. Section 3 gives Maverick and its correctness certificates. Sections 4 and 5 describe the implementation and experiments. Section 6 gives the output-sensitive analysis and scaling scope. Section 7 compares HEnbane and Maverick and discusses the revised parameters. Proofs and additional measurements appear in the appendices.

2 Preliminaries

Groups and words. Unless stated otherwise, this paper targets the original GRAFHEN parameters with $G = S_n$, $n \geq 7$, and $d \geq 2$. Let $A = \{a_1, \ldots, a_d\}$ and $B = \{b_1, \ldots, b_d\}$ be disjoint alphabets, and write A^*, B^* for the free monoids. We write $\mathrm{Stab}(j)$ for a point stabiliser, $C_G(H)$ for the centraliser of $H \leq G$, and $Z(G)$ for the centre. The notation $\mathrm{Inn}(G)$, $\mathrm{Aut}(G)$, and $\mathrm{Out}(G)$ denotes the inner-, full-, and outer-automorphism groups. We use standard permutation-group terminology from [4,14]. Cost statements are in the word-RAM model with $O(\log |G|)$-bit words, sufficient to index group elements and cosets.

Evaluation. Given $\alpha = (\alpha_1, \ldots, \alpha_d) \in G^d$, the evaluation map $\mathrm{ev}_\alpha : A^* \to G$ sends each letter a_i to α_i and extends multiplicatively; $\mathrm{ev}_\beta : B^* \to G$ is defined analogously from β. For mixed words we write $\mathrm{ev}_{(\alpha,\beta)} : (A \cup B)^* \to S_n \rtimes S_n$ for the evaluation sending $a_i \mapsto (\alpha_i, \mathrm{id})$ and $b_i \mapsto (\mathrm{id}, \beta_i)$.

Rewriting systems. A set of rules R over A is *compatible* with α if every rule $u \to v$ satisfies $\mathrm{ev}_\alpha(u) = \mathrm{ev}_\alpha(v)$. The rules generate an equivalence relation $\equiv_R$ on A^*; the quotient $A^*/\equiv_R$ is a monoid that surjects onto $\mathrm{ev}_\alpha(A^*) = \langle \alpha \rangle$. We assume R presents G, meaning the monoid $A^*/\equiv_R$ is isomorphic to G under ev_α; this is a correctness requirement for GRAFHEN.

Shortlex order and canonical form [15, § 2.1]. Fix a total order $a_1 < \cdots < a_d$ on A, and let $\leq_{\mathrm{slex}}$ denote the induced shortlex order on A^* (compare by length; ties by lex). For any $w \in A^*$ the shortlex canonical form

$$\mathrm{NF}_{\mathrm{slex}}(w) \;:=\; \min_{\leq_{\mathrm{slex}}} \{w' \in A^* : w' \equiv_R w\}$$

is well-defined, computable, and satisfies the following (by definition and independently of R):

(i) $\mathrm{NF}_{\mathrm{slex}}$ is a total function $A^* \to A^*$;
(ii) idempotent: $\mathrm{NF}_{\mathrm{slex}} \circ \mathrm{NF}_{\mathrm{slex}} = \mathrm{NF}_{\mathrm{slex}}$;
(iii) shortlex-contractive: $\mathrm{NF}_{\mathrm{slex}}(w) \leq_{\mathrm{slex}} w$;
(iv) evaluation-invariant: $\mathrm{ev}_\alpha(\mathrm{NF}_{\mathrm{slex}}(w)) = \mathrm{ev}_\alpha(w)$;
(v) evaluation-factoring: $\mathrm{ev}_\alpha(w_1) = \mathrm{ev}_\alpha(w_2) \;\Rightarrow\; \mathrm{NF}_{\mathrm{slex}}(w_1) = \mathrm{NF}_{\mathrm{slex}}(w_2)$.

Remark 1 (Comparison with NF_R). The rewriting system's own reduction NF_R (computed by repeatedly applying rules of R) is idempotent and evaluation-invariant, but it need not be evaluation-factoring: in a non-confluent system, two words with the same evaluation can reduce to syntactically distinct irreducible forms. The shortlex canonical form $\mathrm{NF}_{\mathrm{slex}}$, by contrast, is evaluation-factoring by construction.

Cayley graph. The Cayley graph $\mathrm{Cay}(G, \alpha)$ has vertex set G and, for each $g \in G$ and each $i \in \{1, \ldots, d\}$, an edge from g to $g \cdot \alpha_i$ labelled by i; for the diameter we take the graph undirected (equivalently, adjoin the reverse of each edge, so paths may traverse $\alpha \cup \alpha^{-1}$). Its diameter, written $\mathrm{diam}(\mathrm{Cay}(G, \alpha))$, is the smallest integer D such that every $g \in G$ is reachable from id_G by some word of length $\leq D$ over $\alpha \cup \alpha^{-1}$.

Froidure–Pin (FP) enumeration [6]. Given a generating tuple $\alpha = (\alpha_1, \ldots, \alpha_d) \in G^d$, the Froidure–Pin algorithm enumerates the elements of $\langle \alpha \rangle$ in shortlex order: at each step it considers the next shortlex word $w \in A^*$, evaluates $\mathrm{ev}_\alpha(w)$, and either records w as the canonical representative of a new group element or, if $\mathrm{ev}_\alpha(w)$ has already been seen with canonical word $w' <_{\mathrm{slex}} w$, emits the rewriting rule $w \to w'$. Termination is by the word-length bound: at depth $D \geq \mathrm{diam}(\mathrm{Cay}(G, \alpha))$ every group element has been reached. The output is a confluent shortlex rewriting system together with the multiplication table on canonical representatives. GRAFHEN's KeyGen runs an admissibility-filtered variant of this procedure ([8, § 7.1, § A.5]; see Definition 3).

Definition 1 (FP-admissible rewriting system). *A finite rewriting system R over A is* FP-admissible *for α if:*

(i) Evaluation-preserving*: for every $(u, v) \in R$, $\mathrm{ev}_\alpha(u) = \mathrm{ev}_\alpha(v)$.*
(ii) Shortlex-contractive*: for every $(u, v) \in R$, $v <_{\mathrm{slex}} u$.*
(iii) Faithful to G*: the quotient monoid $A^*/\equiv_R$ is isomorphic to G via the map induced by ev_α.*
(iv) Depth-D complete*: there exists a finite $D \in \mathbb{N}$, called the* FP-depth *of R, such that every $w \in A^*$ with $|w| \leq D$ that fails to be $\leq_{\mathrm{slex}}$-minimal in its ev_α-preimage is the LHS of some rule $(w \to w'') \in R$ with $w'' <_{\mathrm{slex}} w$. In GRAFHEN's actual Froidure–Pin output, the relevant left-hand side may occur as a* subword *of w rather than as w itself.*

2.1 Coset tables and the regular representation

We recall coset enumeration following Sims [15, Chapter 4].

Definition 2 (Coset table). *Let $\langle A \mid R\rangle$ be a finite presentation of a group G. A coset table on M cosets is a partial function $T : \{1,\dots,M\} \times (A \cup A^{-1}) \to \{1,\dots,M\}$ satisfying $T(c,x) = c'$ if and only if $T(c',x^{-1}) = c$. A coset table is closed on a relator $r = x_1x_2\cdots x_\ell$ if for every $c \in \{1,\dots,M\}$, tracing c through the letters of r returns to c. As a static object, a closed table gives a permutation action of the presented group; it does not by itself prove that the row stabiliser is trivial.*

Remark 2 (Certification at the target order). Let $R' \subseteq R$ contain only relations valid under a generating tuple α for a known finite group G. There is a surjection $\langle A \mid R'\rangle \twoheadrightarrow G$. Suppose additionally that a proof-carrying Todd–Coxeter trace over the trivial subgroup is replayed and certifies that the presentation has at most $|G|$ elements. If its final closed table has exactly $|G|$ rows, then the surjection is an isomorphism and the table is the right-regular action of G. The replay mechanism is made precise in Lemma 1. This certificate remains valid even when R' was selected heuristically and was not known in advance to present G.

Lemma 1 (Replayable coset-enumeration certificate). *Let $P = \langle A \mid R\rangle$, and let H be either the trivial subgroup or $\langle a_1\rangle$. A Todd–Coxeter run over H can emit a replayable log consisting of* definitions*, relator* deductions*, forced* coincidences*, and row* deletions*. Let T count the table operations and relator letter scans performed by the run, and suppose that at most M cosets are live at any time. The log has $O(T)$ bits, and a deterministic verifier replays it using*

$$O(T + Md) \quad \text{elementary table operations and} \quad O(Md)$$

words of working memory. If the resulting table is complete on n surviving cosets, closes every relator, and, in the case $H = \langle a_1\rangle$, fixes the base coset under a_1, then

$$[P : H] = n.$$

Proof. The verifier maintains the current partial table and a union–find structure. A definition introduces a Schreier-tree edge. For a deduction event it rescans the claimed relator and accepts only if the announced edge is its unique gap. For a coincidence it checks that an existing scan forces the announced merge, propagates the merge, and permits a deletion only after the row is dead. Thus each event is locally checkable.

For soundness, choose freely the tree words t_x naming live cosets. An induction over the replay establishes for every current edge (x, s, y) that

$$t_x s t_y^{-1} \in H\langle\langle R\rangle\rangle.$$

Definitions preserve the invariant freely. A deduction expresses the new edge as a conjugated relator times already justified edges. A coincidence is forced by two

such paths. The set on the right is a subgroup because $\langle\langle R \rangle\rangle$ is normal; when $H = \langle a_1 \rangle$, the base a_1 loop supplies the H factor. At completion, every Schreier generator of the stabiliser of the base coset therefore belongs to $H\langle\langle R \rangle\rangle$. The reverse containment holds from the relator closures and, when needed, the base loop, so the stabiliser is exactly that subgroup. The complete transitive table consequently presents the action on the n right cosets of H in P, proving the claim.

Remark 3 (Trace resources). The replay space is controlled by the peak live-coset count M, not by the final index n. Enumerations can overshoot their final table substantially, so $O(nd)$ space is valid only under an additional bound $M = O(n)$. The log has $\widetilde{O}(T)$ bits when events store row, generator, and relator indices. Trimming it to the ancestor DAG of the final facts can only reduce both the replay work and peak live-row count. For the trivial-subgroup case, the verifier omits the base generator loop; this is the certificate needed for a labelled regular action.

Remark 4 (Group-mode enumeration on GRAFHEN's rules). GRAFHEN publishes a monoid presentation (directed rules $u \to v$). Since the monoid quotient $A^*/\equiv_R$ is isomorphic to a group ($G = S_n$), interpreting each rule as the group relator uv^{-1} yields a group presentation $\langle A \mid \{uv^{-1} : (u, v) \in R\}\rangle$ of the same group G. Group-mode Todd–Coxeter on this presentation with trivial subgroup targets index $|G|$, compared to the monoid-mode index $|G| \cdot |A|$ used in [8, § B]. This is the source of the complexity gap between the present attack and the analysis reported there.

2.2 The targeted GRAFHEN construction

We recall the original S_n construction attacked in this paper. The July 2026 revision changes the platform group to $\mathrm{PSL}_2(343)$ but otherwise retains the same rewriting-system architecture [8, Section 8]; it is discussed separately in Section 7.

Definition 3 (GRAFHEN encryption). *The GRAFHEN scheme at parameters (n, d) with $n \geq 7$, $d \geq 2$ consists of three probabilistic polynomial-time algorithms (*KeyGen*,* Enc*,* Dec*) and a public choice of homomorphic encoding $(E, \mathsf{Enc}_E, L, \pi)$ where E is a finite group, $\mathsf{Enc}_E : K \hookrightarrow E$ is an injective encoding of the message space K (a commutative ring whose additive group embeds in E) ([8, § 2]), $L \leq S_n \rtimes S_n$ is a subgroup, and $\pi : L \to E$ is a surjective homomorphism. The algorithms are:*

$\mathsf{KeyGen}(1^n)$**:**

samples $\alpha = (\alpha_1, \ldots, \alpha_d) \in S_n^d$ and $\beta = (\beta_1, \ldots, \beta_d) \in S_n^d$ uniformly subject to the pairwise generation condition ([8, § 8]: every pair (α_i, α_j) and every pair (β_i, β_j) generates S_n); runs admissibility-filtered Froidure–Pin enumeration ([8, § 7.1, § A.5]) on the alphabets $A = \{a_1, \ldots, a_d\}$, $B = \{b_1, \ldots, b_d\}$, and on the mixed alphabet $A \cup B$, producing rule sets $R_\alpha \subseteq A^* \times A^*$, $R_\beta \subseteq B^* \times B^*$, and $R_{\mathrm{mix}} \subseteq (B \cdot A)^* \times (A^* \cdot B)^*$ whose generic mixed

rule has the form $b_i a_j \to \mathrm{rhs}_{i,j} b_i$ *with* $\mathrm{ev}_\alpha(\mathrm{rhs}_{i,j}) = \beta_i \alpha_j \beta_i^{-1}$ *([8, § 7.2]).*[1] *Outputs the public key* $\mathsf{pk} = (n, d, R_\alpha, R_\beta, R_{\mathrm{mix}}, E, \mathsf{Enc}_E, L, \pi)$ *and secret key* $\mathsf{sk} = (\alpha, \beta)$.

$\mathsf{Enc}(\mathsf{pk}, b)$**:**

for $b \in K$*, samples a pre-image* $x \in L$ *with* $\pi(x) = \mathsf{Enc}_E(b)$*, then a random word* w *over* $A \cup B$ *that evaluates to* x *in* $S_n \rtimes S_n$*, and outputs* w *reduced modulo the public rules. After reduction* w *has the canonical form* $w_A w_B$ *with* $w_A \in A^*$*,* $w_B \in B^*$.

$\mathsf{Dec}(\mathsf{sk}, w)$**:**

computes $g = \mathrm{ev}_{(\alpha,\beta)}(w_A w_B) = (\mathrm{ev}_\alpha(w_A), \mathrm{id}) \cdot (\mathrm{id}, \mathrm{ev}_\beta(w_B)) \in S_n \rtimes S_n$*, applies the public projection* $\pi : L \to E$*, and outputs* $\mathsf{Enc}_E^{-1}(\pi(g)) \in K$.

The original recommended parameters were $n = 11$*,* $d = 5$*; these are the parameters of the released challenges and of our end-to-end experiments. They are no longer the recommendation in the July 2026 revision [8, Section 8].*

Remark 5 (Choice of (E, L, π)*).* The recommended choice ([8, § 8]) takes $L = S_n \rtimes S_8$ with $E = S_n$ and $\pi = \pi_0 \circ f$, where $f(x, y) = xy$ is the multiplication map on $S_n \rtimes S_8$ and π_0 is a fixed homomorphism $S_n \to E$. Other partition choices (E, L, π) with $E \rtimes H \subseteq S_n \rtimes S_n$ are admissible ([8, § A.1]); Maverick is partition-agnostic through Stage 4(i) (Section 5.2).

Definition 4 (Equivalent decryption capability). *Given a GRAFHEN public key* pk*, an algorithm* Dec^* *is a* functionally-equivalent decryption capability *if for every* $b \in K$ *and every honest ciphertext* $c \leftarrow \mathsf{Enc}(\mathsf{pk}, b)$*,* $\Pr[\mathsf{Dec}^*(c) = b] = 1 - \mathrm{negl}(n)$ *over the randomness of* Enc *and any internal randomness of* Dec^*.

Remark 6 (Why the conjugator is unnecessary). A simultaneously conjugate tuple $\alpha' = \sigma\alpha\sigma^{-1}$, $\beta' = \sigma\beta\sigma^{-1}$ evaluates every ciphertext to the σ-conjugate of its secret-key evaluation. The attacker generally cannot recover σ, because the secret tuple (α, β) is not public. Instead, public chosen-plaintext encryptions transport the decryption fibres into the recovered coordinate system. For the binary instances, zero samples generate the transported kernel and a one sample identifies its other coset; Proposition 2 formalises this calibration. This recovered tuple and calibrated classifier are the equivalent decryption capability considered here.

GRAFHEN's Froidure–Pin construction [8, § 3.4] produces rules whose depth can be bounded below:

Proposition 1 (FP-depth, for a complete enumeration). *Let* R *be the complete rule set produced by running Froidure–Pin until all elements of* $\langle\alpha\rangle = S_n$ *have been enumerated, and let* D_R *be the maximum depth reached. Then* $D_R \geq \mathrm{diam}(\mathrm{Cay}(S_n, \alpha))$.

[1] This is the pre-completion format of [8, § 7.2]. Knuth–Bendix completion on $R_\alpha \cup R_\beta \cup R_{\mathrm{mix}}$ may produce mixed rules with multi-letter β-affixes; this case is handled with identical asymptotic cost (Section A).

Proof. Every $g \in S_n$ has then appeared with a positive representative of length at most D_R. Such a representative is also a word over $\alpha \cup \alpha^{-1}$, so the undirected Cayley diameter is at most D_R.

GRAFHEN may publish a pseudo-bounded filtered subsystem rather than the complete FP output. Proposition 1 therefore describes the complete enumeration, not a guarantee that every selected sub-corpus presents S_n; the target-order certificate of Remark 2 is what the attack checks.

Remark 7 (FHE compactness). Homomorphic operations lengthen ciphertexts. To remain compact, public reduction must be available at every depth up to the operational length, which is condition (iv) of Definition 1.

3 The Attack

Maverick processes the public rule sets R_α, R_β, and R_{mix} from Definition 3. The implementation selects the K shortest per-side rules for performance, but selection is not trusted: the theorem-level certificate consists of a target-order table and a replayable trace. The algebra below proves that such a completed run is correct; Section 6.6 separately states the hypotheses under which a selected corpus is expected to complete.

3.1 Stage 1: certified regular-action reconstruction

Interpret a directed rule $u \to v$ as the group relator uv^{-1}. On each alphabet, run Todd–Coxeter for the trivial subgroup and target the known order $N = |S_n| = n!$.

Lemma 2 (Target-order certificate). *Let $\alpha \in S_n^d$ generate S_n, and let $R' \subseteq R_\alpha$ be any set of public rules. If Todd–Coxeter on $Q = \langle A \mid \{uv^{-1} : (u, v) \in R'\}\rangle$ returns a closed table with exactly $n!$ rows and a proof-carrying trace certifying the trivial-subgroup enumeration, then $Q \cong S_n$ under $a_i \mapsto \alpha_i$, and the table is the right-regular action of S_n.*

Proof. Every relation in R' is valid under α, and α generates S_n, so evaluation induces a surjection $Q \twoheadrightarrow S_n$. A replay-verified trivial-subgroup trace proves $|Q| \le n!$, while the surjection proves $|Q| \ge n!$. A surjection between finite groups of equal order is an isomorphism, and the table is the regular action.

The same argument applies to R_β. Thus Stage 1 produces certified tables T_α, T_β or returns "inconclusive"; workspace exhaustion or failure to reach the target is never interpreted as a key.

3.2 Stage 2: constructive recognition

A regular table labels group elements opaquely. Constructive recognition turns it into the natural degree-n action.

Lemma 3 (Generator recovery from a certified table). *Given a certified regular table for S_n, $n \geq 7$, constructive recognition returns a tuple $\alpha' \in S_n^d$ such that*

$$\alpha'_i = \sigma_\alpha \alpha_i \sigma_\alpha^{-1} \qquad (1 \leq i \leq d)$$

for some $\sigma_\alpha \in S_n$. The output is certified by checking the generator action and the order of a point stabiliser.

Proof. Constructive recognition of symmetric groups in an explicit representation is standard [1,2]. It produces the action on the cosets of a point stabiliser of order $(n-1)!$. For $n \neq 6$, every automorphism of S_n is inner and every index-n subgroup is a point stabiliser [4, Theorem 5.2A]. Hence the recovered action is simultaneously conjugate to the secret one. Here $n \geq 7$, so the exceptional outer automorphism of S_6 does not arise.

Our implementation uses the specialised candidate scan of Section A.7. It does not accept an element merely because its order is n: for composite n, such an element need not be an n-cycle. Acceptance requires the exact point-stabiliser and action certificates of Lemma 3. Applied independently, Stage 2 gives

$$\alpha' = \sigma_\alpha \alpha \sigma_\alpha^{-1}, \qquad \beta' = \sigma_\beta \beta \sigma_\beta^{-1}.$$

3.3 Stage 3: alignment with mixed relations

The two tuples are expressed in unrelated coordinate systems. The canonical mixed rules include, for every $i, j \in \{1, \ldots, d\}$,

$$b_i a_j \longrightarrow w_{i,j} b_i, \qquad \mathrm{ev}_\alpha(w_{i,j}) = \beta_i \alpha_j \beta_i^{-1}. \tag{1}$$

They determine the relative coordinate change.

Lemma 4 (Trivial joint centraliser). *Let G be centreless and let each of $\alpha, \beta \in G^d$ generate G. The simultaneous centraliser of (α, β) is trivial. This holds for the original GRAFHEN keys in S_n, because KeyGen requires every pair on each side to generate S_n.*

Proof. The simultaneous centraliser is contained in $C_G(\langle\alpha\rangle) = C_G(G) = Z(G) = 1$.

Theorem 1 (Certified equivalent-key recovery). *Let $n \geq 7$ and suppose each secret per-side tuple generates S_n. Assume Stage 1 returns certified $n!$-row tables on selected subsets $R_\alpha^* \subseteq R_\alpha$ and $R_\beta^* \subseteq R_\beta$. Then Maverick recovers, using the d^2 canonical mixed rules of eq. (1), a tuple (α', β'') for which*

$$\alpha' = \sigma \alpha \sigma^{-1}, \qquad \beta'' = \sigma \beta \sigma^{-1}$$

for a common, generally unknown $\sigma \in S_n$. The output can be verified against all public rules. Apart from the two coset enumerations, the algorithm uses polynomial time in the encoded table size plus at most $n!$ alignment tests.

Proof. Lemmas 2 and 3 give $\alpha' = \sigma_\alpha \alpha \sigma_\alpha^{-1}$ and $\beta' = \sigma_\beta \beta \sigma_\beta^{-1}$. The correct relative conjugation is $\rho^* = \sigma_\alpha \sigma_\beta^{-1}$; setting $\beta'' = \rho^* \beta' (\rho^*)^{-1}$ puts both tuples in the σ_α frame.

To recover ρ^*, enumerate $\rho \in S_n$ and test all equations eq. (1) under $(\alpha', \rho\beta'\rho^{-1})$. Existence follows from the displayed value of ρ^*. For uniqueness, conjugate an accepting equation back by σ_α^{-1} and set $\mu = \sigma_\alpha^{-1} \rho \sigma_\beta$. For fixed i and every j, the equations imply

$$\beta_i^{-1} \mu \beta_i \mu^{-1} \in C_{S_n}(\alpha_j).$$

Since all j occur and $\langle \alpha \rangle = S_n$, this commutator is the identity. Thus μ centralises every β_i, and $\langle \beta \rangle = S_n$ gives $\mu = 1$. Hence $\rho = \rho^*$. Verification on the complete public corpus is direct.

Theorem 2 (Centreless two-side recovery template). *Let G be a finite centreless group of known order. Let $\alpha, \beta \in G^d$ generate G, and suppose public per-side rule subsets yield replay-certified closed trivial-subgroup tables of order $|G|$. Assume a constructive recognizer converts each table into a tuple in a fixed explicit model of G, up to an automorphism of G. If the public mixed relations specify all conjugation equations*

$$\beta_i \alpha_j \beta_i^{-1} \qquad (1 \le i, j \le d),$$

then enumerating $\mathrm{Aut}(G)$ *uniquely aligns the two tuples into a common automorphic image of (α, β).*

Proof. The target-order argument of Lemma 2 identifies both tables with the regular action of G. By the recognizer hypothesis, the recovered tuples are $\phi_\alpha(\alpha)$ and $\phi_\beta(\beta)$ for automorphisms $\phi_\alpha, \phi_\beta \in \mathrm{Aut}(G)$. The automorphism $\phi_\alpha \phi_\beta^{-1}$ satisfies every mixed equation.

Let δ be another accepting automorphism and set $\theta = \phi_\alpha^{-1} \delta \phi_\beta$. Pulling the mixed equations back to G shows, for every i, j, that $\beta_i^{-1} \theta(\beta_i)$ centralizes α_j. Since $\langle \alpha \rangle = G$, it lies in $Z(G) = 1$. Hence θ fixes every β_i; since $\langle \beta \rangle = G$, $\theta = 1$, proving uniqueness.

Corollary 1 (Common-coordinate tuple). *After Stage 3, the recovered tuple is a simultaneous conjugate of the secret tuple. Recovering the conjugator σ is neither claimed nor needed.*

3.4 Stage 4: chosen-plaintext calibration

GRAFHEN encryption is public, so the attacker can generate labelled ciphertexts. Let f denote the group homomorphism applied before the final binary decoding, and let H be the kernel corresponding to plaintext zero. For the released binary instances, evaluations of zero encryptions lie in H, while evaluations of one encryptions lie in a disjoint left or right coset represented by some t.

Lemma 5 (Calibration in the recovered frame). *Let (α', β'') be the common-coordinate tuple of Corollary 1. Suppose evaluated zero samples generate $H' = \sigma H \sigma^{-1}$, and let t' be the evaluation of one labelled one-ciphertext. Then H' and the appropriate coset $t'H'$ (or $H't'$) classify every honest binary ciphertext.*

Proof. Evaluation under the recovered tuple conjugates every secret-frame evaluation by the same σ. It therefore maps H to H' and the one-fibre tH (respectively Ht) to $t'H'$ (respectively $H't'$). Conjugation preserves disjointness and coset membership.

Proposition 2 (Equivalent decryption capability). *Under the sampling condition of Lemma 5, the tuple recovered by Theorem 1, together with the calibrated subgroup and coset, is an equivalent decryption capability in the sense of Definition 4. It uses only the public key and public chosen-plaintext encryptions.*

Subgroup generation and membership are computable by Schreier–Sims algorithms. The condition that zero samples generate the transported kernel is testable by monitoring subgroup growth and validating on fresh labelled samples. In our experiments it holds for every instance, and each decryptor is checked on 20,000 labelled ciphertexts (Section 5.3).

Remark 8 (Single-TC variant). When the mixed rules retain the canonical form eq. (1), one may run Stage 1 only on R_α and recover the second tuple directly from the mixed relations. We use the two-side implementation because it applies to the released rule formats without relying on that normal form.

4 Implementation

We implemented the attack with a purpose-built coset enumerator based on ACE [10], extended with the algorithmic improvements of Section A, layered with admissibility-filtered Froidure–Pin rule generation following GRAFHEN's published procedure ([8, § A.5]). All measurements in Section 5 use this implementation.

4.1 Rule-generation procedure

GRAFHEN's key generation [8, § A.5] uses a custom Froidure–Pin enumeration with an admissibility filter applied either *after* enumeration (post-selection: retain only admissible rules from the complete canonical set) or *during* enumeration (pre-selection: promote non-admissible reductions to new canonical forms, producing a larger corpus). Post-selection yields $\sim 10^4$–10^7 rules at $n \in \{9, 11\}$; pre-selection yields $\sim 10^5$–10^7. GRAFHEN's reported corpus sizes ("about 20 million rules per side at $n = 11$" [8, § 8]) match pre-selection. We use post-selection in our primary measurements (via libsemigroups [3]) and pre-selection as a cross-check (via [9]). In both cases, selected prefixes reach closure-checked tables at the same target coset count. A theorem-level certificate additionally requires the replay trace of Lemma 2.

The pipeline was calibrated end-to-end at $n = 9$, where a known test key allows an additional secret-tuple check. The corpus-prefix sweep, including the observed transition from workspace exhaustion to completed closure checks, is reported in Section B. It motivates the per-corpus measurement procedure of Section 4.2; it is not used as a scaling prediction.

4.2 Rule-count selection

The attacker feeds only the K shortest rules to Todd–Coxeter. The elapsed-time-optimal K balances two competing effects: too few rules cause workspace overflow from insufficient coincidences, while too many rules inflate the per-coset scan cost. We choose K empirically for each corpus and accept it only when the target-order closure check succeeds. At $n = 11$, the operating points are $K = 670{,}000$ for the post-admissibility corpus and $K = 575{,}000$ for the pre-admissibility corpus; both yield closed 11!-row tables with $M^*/|G| \approx 6$.

In the direct-coverage surrogate, $K\bar{L} \geq |G|$ is the elementary counting lower bound, while a coupon-collector calculation produces a model-based sufficient threshold with logarithmic overhead. Neither statement is a rigorous lower or upper bound for the actual enumeration; the limitations are detailed in Section C.

5 Empirical results

The completed recovery measurements concern the original recommended construction $(n, d) = (11, 5)$. Smaller instances provide implementation calibration.

5.1 Original recommended parameters

For twelve fresh keys produced by the original S_{11} GRAFHEN procedure, selected per-side corpora yielded closed 11!-row tables and a common-coordinate tuple satisfying the public mixed relations. The median end-to-end time was 552 seconds, with a range of 510–599 seconds, on an Apple M3 Pro. Median peak RSS was 11.28 GB. The selected prefixes recover the tuple; the complete public corpora verify it. The dominant work was the two Todd–Coxeter enumerations; constructive recognition and alignment were included in the stated wall time.

measurement	S_{11} experiment	S_9 calibration
keys / runs	12	12
median elapsed time	552 s	3 s
elapsed-time range	$[510, 599]$ s	$[2, 3]$ s
median peak RSS	11.28 GB	0.16 GB
median peak-row ratio $M^*/\|G\|$	5.94	9.39

Measured completed runs. The peak-row ratio is a diagnostic statistic, not a proved scaling invariant.

Every completed run was checked in three ways: the per-side tables passed the target-order closure check; the aligned tuple satisfied the complete public rule corpus; and the resulting decryptor was evaluated on independent labelled ciphertexts. These experiments did not emit replayable enumeration traces, so they are closure-checked recoveries rather than trace-certified runs. A failed enumeration or check was not counted as a recovered key.

5.2 Chosen-plaintext decryptor validation

We calibrated the recovered-frame kernel and bit-one coset using public encryptions of both bits, as in Lemma 5. On two author-generated original-format instances at $n = 11$, with $d = 5$ and $d = 6$, respectively, the calibrated classifier made no errors on the available 2,500 challenge ciphertexts. These runs also passed validation on 20,000 labelled ciphertexts per instance.

This is an equivalent-decryption result. The classifier operates in the recovered common coordinate frame and uses the sampling condition in Lemma 5, followed by explicit held-out-label checks.

5.3 Published challenge instances

We also evaluated the implementation against the released original GRAFHEN challenge corpus. It completed the target-order closure and complete-rule checks for the published S_{11} instance, and the calibrated decryptor correctly classified all 20,000 supplied labelled ciphertexts. The reported one-side enumeration used two million selected rules, took 391.5 seconds, and reached a peak ratio $M^*/|G| = 5.32$ with 12.1 GB peak RSS.

The labelled-set result establishes operational correctness for that challenge format. Scaling beyond the measured corpora and the revised $\mathrm{PSL}_2(343)$ parameters are addressed separately in Section 6.6 and Section 7.

6 Runtime analysis and scaling scope

The correctness statement is exact: a replay-verified target-order table triggers Lemma 2, and the subsequent stages are certified by public checks. The cost of obtaining that certificate depends on the selected corpus and the peak table size. General Cayley-diameter results do not determine either quantity for GRAFHEN's filtered corpus.

6.1 Output-sensitive enumeration cost

Let K be the number of selected relators, let $\bar{L}$ and L be their mean and maximum lengths, and let M^* be the maximum number of live rows created during an enumeration. The standard HLT analysis gives an output-sensitive bound:

$$T_{\mathrm{TC}} = O(M^* K L)$$

word operations, up to the representation and relator-indexing costs of the implementation [15]. The mixed HLT/Felsch strategy used here has the same conservative asymptotic bound and can improve the closure pass in practice (Theorem 6).

Theorem 3 (Output-sensitive runtime bound). *For any selected corpus on which Stage 1 returns a certified regular table, Maverick runs in*

$$O\big(T_{\mathrm{TC}}(R_\alpha^*) + T_{\mathrm{TC}}(R_\beta^*) + \mathrm{poly}(|G|, n, d)\big)$$

operations, plus at most $|S_n|$ tests of the d^2 mixed relations for alignment. The hidden constants and the peak sizes are instance dependent.

Proof. Stage 1 has the stated output-sensitive cost. The target-order certificate, constructive-recognition certificate, complete-rule verification, and Schreier–Sims calibration are polynomial in their explicit inputs. The alignment loop enumerates at most $|S_n|$ candidates and performs a constant number of permutation-word evaluations per candidate when d is fixed.

6.2 What diameter results do and do not imply

For context, Helfgott–Seress prove a quasipolynomial diameter bound for every generating set of S_n [11], and Helfgott–Seress–Żuk give a polynomial diameter bound for random generators [12].

Theorem 4 (Helfgott–Seress). *For every generating set A of S_n or A_n,*

$$\mathrm{diam}(\mathrm{Cay}(S_n, A)) \le \exp\big(O((\log n)^4 \log\log n)\big).$$

Theorem 5 (Helfgott–Seress–Żuk). *For a uniformly random generating pair of S_n (or A_n), the Cayley diameter is $n^{O(1)}$ with probability $1 - o(1)$.*

Corollary 2 (Diameter for a random tuple). *If a random d-tuple contains a generating pair satisfying Theorem 5, then adding the remaining generators cannot increase the Cayley diameter. Conditioning a tuple on GRAFHEN's pairwise-generation test requires care, but preserves any $1 - o(1)$ property whose failure probability remains $o(1)$ after that conditioning.*

These theorems concern the generator graph. They neither prove a spectral gap for that graph nor describe the distribution of Froidure–Pin rules retained by GRAFHEN's admissibility filter. In particular, they do not bound K, L, or M^* for the K shortest public rules. The required corpus and peak-growth assumptions are stated explicitly below.

6.3 Selected-corpus hypotheses

Definition 5 (Corpus-convergence hypothesis). *For a distribution of GRAFHEN keys and a specified rule-selection procedure, there are functions K_{conv} and $\varepsilon_n = o(1)$ such that Todd–Coxeter on the selected K_{conv} rules returns a certified replay-verified $|G|$-row table with probability at least $1 - \varepsilon_n$.*

Definition 6 (Peak-growth hypothesis). *On the same distribution and selected corpus, the peak table size satisfies $M^* = |G|(\log |G|)^{O(1)}$ with probability at least $1 - \varepsilon_n$.*

The random-walk coupon-collector calculation in Section C motivates a candidate scale

$$K_{\mathrm{conv}} = O\left(\frac{|G|(\log |G|)^2}{\lambda \bar{L}}\right),$$

provided the selected corpus behaves like suitably spaced random walks and direct coverage is a reliable proxy for enumeration convergence. Neither premise has been proved for the actual FP-admissible corpus. We therefore use this expression only as a model, not as a theorem about GRAFHEN.

6.4 Implementation certificates

Theorem 6 (Target-aware closure). *Consider a Todd–Coxeter enumeration with known target order N. If a tentative N-row table is followed by a complete closure pass and accepted only when no new coincidence arises, then the accepted table is closed. With a replayable proof-carrying enumeration trace, it certifies the regular action. The choice of the stability threshold used before the closure pass affects performance but not correctness.*

Proof. If closure finds a coincidence, enumeration resumes and no output is accepted. If closure finds none, every relator scan is closed. Together with the target-order check and a replay-verified trace, Lemma 2 applies.

Theorem 7 (Certified constructive recognition). *The implementation-level recognition procedure is sound whenever it returns: it verifies an index-n point stabiliser and the induced degree-n action. Standard constructive-recognition algorithms supply a complete alternative for S_n [1].*

6.5 Conditional scaling statement

Theorem 8 (Conditional scaling). *Assume Definitions 5 and 6 and suppose $K_{\mathrm{conv}} = \widetilde{O}(|G|/\bar{L})$. Then the Stage 1 cost is $\widetilde{O}(|G|^2)$, and Maverick has the same bound up to the cost of alignment and constructive recognition.*

Proof. Substitute $M^* = \widetilde{O}(|G|)$ and $K_{\mathrm{conv}}\bar{L} = \widetilde{O}(|G|)$ into the output-sensitive bound.

Remark 9 (No asymptotic security frontier). The measurements through $n = 11$ support neither a numerical 2^{128} crossover nor a statement about all larger degrees. Such claims would require proved constants for the selected-corpus and peak-growth hypotheses. The paper's concrete security conclusion is limited to the instances actually recovered in Section 5.

Remark 10 (Key-generation cost). The cost of producing and storing a public rewriting corpus is an engineering constraint, not a universal bound on permissible parameters. The $\mathrm{PSL}_2(343)$ measurement in Section E.2 illustrates this distinction.

6.6 Scaling observations and open questions

The S_{11} experiments are compelling evidence of a practical break, but they are not a proof of a scaling law. This section records the two regularities observed in the data and the reason neither is promoted to a theorem.

Conjecture 1 (Stable peak ratio). For the original S_n GRAFHEN generator and the selected-corpus strategy used by Maverick, $M^*/|G|$ remains bounded with high probability as n grows.

Conjecture 2 (Sub-direct-coverage convergence). For the same corpus model, the number of selected rules needed for a certified table is below the direct-coverage scale $|G|/\bar{L}$ by a factor comparable to a Cayley-graph distance scale.

The data are consistent with both statements: the median peak ratios at $n = 8, 9, 10, 11$ are $27.4, 9.4, 7.4, 5.94$, and the selected corpus at $n = 11$ converges well below $|G|/\bar{L}$. The public challenge corpus instead needs about $|G|/\bar{L}$ rules while retaining a small peak ratio. This variation is useful evidence that corpus structure matters; it also rules out treating a single fitted constant as a universal prediction.

6.7 Why the coupon-collector calculation is conditional

For independent or suitably mixing random walks, sampling every $O(\lambda^{-1} \log |G|)$ steps gives the familiar coupon-collector scale

$$K = O\left(\frac{|G|(\log |G|)^2}{\lambda \bar{L}}\right).$$

The selected GRAFHEN corpus is neither an independent sample of walks nor a uniform sample from all words. Froidure–Pin canonicalisation, admissibility, length selection, and the enumeration strategy all introduce dependencies. Moreover, direct visitation of every group element is not equivalent to Todd–Coxeter convergence: coincidences can propagate information into unvisited portions of the table. The calculation is therefore a motivation for Definition 5, not a proof of it.

6.8 Empirical scaling

Across the measured degrees, Stage 1 wall time increases far more slowly than the conservative worst-case table bound. We report the measurements as descriptive data rather than fitting a security frontier. They establish that the implementation is practical at the original recommended parameters; they do not establish a limiting exponent or predict the cost of a new parameter set.

6.9 Open problem

A proof-quality scaling result must connect the distribution of the published, filtered rules to a particular fair coset-enumeration strategy. It must control both the probability of reaching a target-order certificate and the peak number of speculative rows. Spectral information about a uniformly sampled Cayley walk is insufficient on its own. The revised $\mathrm{PSL}_2(343)$ parameters make this question more pressing, because their group order is within the range of the present per-side enumeration machinery while target-size public-rule recovery has not yet been measured end to end.

7 Concurrent work and the revised parameters

7.1 Relationship with HEnbane

Dumezy's concurrent HEnbane attack [5] and Maverick have the same cryptographic endpoint: both recover an equivalent representation and use public known-plaintext information to obtain a decryptor. Their reconstruction methods differ.

HEnbane first derives short consequences through bounded congruence closure and right-hand-side cancellation. It then reconstructs a degree-n permutation tuple by conflict-driven search over partial permutation tables, and uses mixed relations to obtain the second tuple in the semidirect construction. Its complete fallback has a degree- and generator-parameterised exponential bound, and its paper deliberately makes no distributional claim for untested parameters.

Maverick treats the per-side rules as a group presentation. Its first stage reconstructs a regular action by target-order coset enumeration and then recognises the natural permutation action. Its dominant resources are the group order, the selected corpus, and the peak coset table. Dumezy's consequence mining is a plausible preprocessing component for a future hybrid.

Proposition 3 (Representation-model separation). *Let G be a finite group generated by a labelled d-tuple $S = (s_1, \ldots, s_d)$, and let R be a fixed set of word relations in S. The presentation*

$$P_R = \langle S \mid R \rangle$$

and every deterministic Todd–Coxeter run on P_R depend on R, its ordering, and the enumeration policy, but not on a faithful permutation representation of G. In

contrast, a complete search whose candidates are labelled degree-r permutation tuples has ambient search space $(S_r)^d$, of size $(r!)^d$; for two independent tuples it has ambient size $(r!)^{2d}$.

Proof. Faithfulness makes word equality representation-independent:

$$\rho(\mathrm{ev}_S(u)) = \rho(\mathrm{ev}_S(v)) \quad \Longleftrightarrow \quad \mathrm{ev}_S(u) = \mathrm{ev}_S(v)$$

for every pair of words u, v. Thus the relators defining P_R are the same under every faithful representation. Todd–Coxeter manipulates only these relators, their scan order, and entries in a coset table, proving the first claim. The second follows by counting the $r!$ possible labelled images of each of the d generators; the two-side count is the product of the two independent candidate spaces.

Proposition 3 is a comparison of algorithmic models, not a lower bound on HEnbane or a completion guarantee for Maverick. Changing from S_{11} to $\mathrm{PSL}_2(343)$ changes the abstract group and the public corpus as well as the representation degree. The proposition isolates the narrower point: HEnbane's permutation-table search explicitly carries the degree parameter, whereas Maverick's target-order reconstruction is driven by the relation presentation and the target group order.

Experimental comparison. Both attacks recover equivalent keys and decrypt every released GRAFHEN challenge. The available measurements are not a controlled head-to-head benchmark. HEnbane reports approximately 4 GB peak memory on a 32-thread AMD Ryzen 9 9950X; its degree-11 runs are seed-dependent, ranging from about 10 seconds to several hours, with some attempts reaching a 4-hour budget without completion [5]. For one public-key variant, its restart study reports four successes among 52 independent 10-minute trials. Maverick recovers all released challenges and, on 12 independently generated original-format S_{11} keys, has median wall time 552 s, range 510–599 s, and median peak RSS 11.28 GB on an Apple M3 Pro. The corpus-selection policies, hardware, thread counts, and measured workloads differ, so these figures do not justify a speed ranking. They instead illustrate complementary tradeoffs: HEnbane has lower reported memory and randomized-search variability, whereas Maverick has stable measurements on the independently generated keys studied here.

7.2 Revised parameters

The July 2026 GRAFHEN revision replaces the original S_{11} platform group by $\mathrm{PSL}_2(343)$ in its degree-344 projective action [8, Section 8]. This directly addresses attacks whose candidate space is governed by the displayed permutation degree. It does not by itself resolve the per-side presentation-recovery approach used by Maverick, whose first stage is governed by the public relations, the target group order, and the peak coset table. The authors' stated worst-case estimate for HEnbane is $O(2^{dn})$ for d generators in degree n. Thus the degree-170, $d = 5$ setting $\mathrm{PSL}_2(169)$ already corresponds to their $O(2^{850})$ threshold, whereas the recommended degree-344 setting gives $O(2^{1720})$. This is an extrapolation of

their search bound, not a measured runtime for HEnbane. At the target group, Maverick's post-enumeration recovery from a generated regular action takes 35.0 s at 1.92 GiB peak memory, and its permutation-side check takes 0.83 s at 85 MB. Thus the revised representation degree is not a practical barrier to this part of Maverick. A generated full-FP corpus suite, including a two-side $\mathrm{PSL}_2(169)$ instance, recovers and decrypts end to end. At $q = 169$, the critical path is 103 s at 0.93 GiB peak memory and decrypts all 5,000 held-out ciphertexts without error. These are unfiltered Froidure–Pin controls, but the $k = 2$ admissibility-filtered corpus at the same parameter also reaches target order. What remains unmeasured is admissibility-filtered public-rule recovery at the revised target $\mathrm{PSL}_2(343)$. Appendix E contains the group-specific algorithms, theorems, and detailed measurements.

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A Optimisation details

This appendix records implementation choices used in the reported experiments. They are performance heuristics, not ingredients of the correctness proof or of the conditional scaling statement. The reported runs use target-order closure checks; a replayable trace is the additional artifact required for the certificate of Lemma 2.

A.1 Rule ordering and selection

Rules are processed in shortlex order. We select a prefix of the shortest public rules because, on the measured corpora, this reduces elapsed time and the number of speculative rows relative to a random order. The selected prefix is an experimental parameter: failure to reach a closed target-order table is reported as inconclusive, rather than as recovery.

A.2 Corpus-size sweep

For the $n = 9$ corpus, the sweep described in Section B.2 has three observed regimes: small prefixes exhaust the assigned workspace, an intermediate range gives the lowest elapsed time, and larger prefixes increase scan work. This is a description of that corpus on the reported platform; it is not a convergence theorem or a parameter-selection rule for other groups.

A.3 Target-aware closure

The enumerator alternates HLT-style definitions with Felsch-style deduction processing. When the live-row count first reaches the known target order and remains stable for a configured interval, it runs a complete closure pass. Only a closure pass that produces neither a new definition nor a coincidence is accepted as a candidate table. Thus the interval changes only the amount of work performed before the closure check, as formalised in Theorem 6.

A.4 Per-side enumeration

For the original semidirect-product construction, the implementation enumerates the two public per-side presentations separately. The mixed rules are then used to align the two reconstructed natural actions as in Theorem 1. This avoids treating the public presentation as a direct enumeration problem of target size $|G|^2$; it does not assert that an arbitrary semidirect-product presentation decomposes in this way.

A.5 Mixed-rule alignment

The implementation uses the two-table procedure of Theorem 1. A single-table shortcut is possible only when the public mixed rules retain the canonical form of eq. (1); it is not used for the released formats. Alignment is accepted only after the resulting common-coordinate tuple satisfies the complete public rule corpus.

A.6 Engineering details

The reported implementation indexes rules by their first applicable column, stores defined columns compactly per row, and uses batched deduction queues. Short rules have a specialised scan path; long rules use the generic scanner. The two independent per-side enumerations may execute concurrently. These choices affect elapsed time and memory use but not the theorem-level certificate obtainable by logging a completed enumeration.

A.7 Certified constructive recognition

The implementation searches the completed regular table for candidate structures that yield an index-n subgroup, constructs the induced action on its cosets, and verifies the recovered generator images. Under the theorem-level replay condition, this table is certified. A candidate is not accepted solely from an element-order test: in particular, an element of order n need not be an n-cycle when n is composite. The acceptance conditions are exactly those in Lemma 3 and Theorem 7. Standard constructive recognition for symmetric groups provides an independent complete procedure for this step [1,2].

A.8 Unsuccessful variants

On the experimental corpora, Felsch-only enumeration and a low-index point-stabiliser enumeration did not improve the measured runtime. We also did not rely on Knuth–Bendix completion: non-termination or failure to complete is irrelevant to the target-order certificate used here.

B Calibration experiments

B.1 Calibration run at $n = 9$

We used a known S_9 test key to exercise the complete implementation before the S_{11} experiments. For the seed reported in the implementation archive, the GRAFHEN rule generator produced 45,075 admissible rules and the shortest 10,000 rules yielded a closed table with the target $9! = 362{,}880$ rows. Constructive recognition, mixed-rule alignment, and complete-corpus verification then succeeded.

On an Apple M3 Pro using eight worker threads, the run took 2.5 seconds and had peak resident memory below 0.2 GB. These numbers are a debugging and reproducibility calibration, not an extrapolation to larger degrees. The known test key also lets us check that the recovered tuple is simultaneously conjugate to the generation tuple; this check is stronger than, and separate from, the public verification used in an attack.

B.2 Observed corpus-size effect

At $n = 9$, prefixes shorter than roughly 2,000 rules exhausted the configured workspace, whereas a broad range near 7,500–10,000 rules gave the lowest elapsed time. Adding substantially more rules increased scan time. This motivated the per-corpus prefix sweep used for the S_{11} measurements. A failed sweep point is recorded as an exhausted resource budget, not evidence that no successful prefix exists.

The experiment illustrates that rule count alone is not an adequate model: rule lengths, dependencies induced by Froidure–Pin enumeration, and the enumeration strategy all affect convergence. The coupon-collector surrogate in Section C is included only to make this modelling gap explicit.

C Coverage model and the open convergence problem

This appendix records the random-walk surrogate used to motivate rule selection. It is not a proof that the selected GRAFHEN corpus converges.

Direct-coverage surrogate. Suppose, only for this paragraph, that each selected relator contributes a walk of mean length $\bar{L}$ and that successful enumeration requires every group element to be visited directly. Then direct coverage has the deterministic counting lower bound

$$K\bar{L} \geq |G|.$$

This is a lower bound for the surrogate, not for Todd–Coxeter itself: coincidence propagation can merge rows associated with elements that no relator walk visits directly.

Proposition 4 (Coupon-collector surrogate). *Let sampled relator-prefix positions form an adapted sequence whose conditional law is within $|G|^{-2}$ in total variation of uniform on G, and suppose that direct coverage is sufficient for the enumeration strategy under study. Then*

$$K \geq \frac{C}{\lambda} \frac{|G|(\log |G|)^2}{\bar{L}}$$

is sufficient for direct coverage with probability $1 - o(1)$, where λ controls the mixing spacing and C is an absolute constant.

Proof. Space samples by $t = \Theta(\lambda^{-1} \log |G|)$ steps. Conditional near-uniformity couples the resulting sequence to uniform draws with total error $o(1)$ at the stated sample count. The usual union bound gives coverage after $O(|G| \log |G|)$ such draws. Each relator supplies $\Theta(\bar{L}/t)$ spaced positions, yielding the bound.

Why this does not prove Definition 5. Spectral mixing applies to a specified random walk, whereas the actual corpus is formed by canonicalisation, admissibility filtering, and a deterministic shortest-first selection. Those operations do not preserve the conditional endpoint distribution assumed above. Further, the direct-coverage condition is only a surrogate for a table process that can both propagate and delay coincidences. The experiments show that the surrogate is useful for choosing K; they do not establish its hypotheses.

Interpretation of the sweep. At $n = 9$, the selected-corpus sweep has a sharp feasibility boundary near $K = 2{,}000$ and a time minimum near $K = 8{,}500$ (Section B.2). We use the minimum as an engineering choice for that corpus. It is not a theorem-derived optimum and should not be transferred unchanged to another group, generator distribution, or rule filter.

D Spectral gap measurements

This appendix records an auxiliary measurement of the generator graphs used in small calibration runs. It is not used to prove convergence of the selected Froidure–Pin corpora, and it is not an assumption of the certified recovery result.

For random pairwise-generating 5-tuples in S_n, $5 \leq n \leq 10$, we formed the lazy random-walk operator on $\mathrm{Cay}(S_n, \alpha \cup \alpha^{-1})$ and estimated its second eigenvalue numerically. Across thirty samples at each degree, the observed lazy-walk gaps had median approximately 0.19; the smallest observed value was 0.11.

These values are consistent with rapid mixing of the *generator* walks at the sampled degrees. They do not imply a spectral-gap theorem, and do not identify the distribution of rules surviving canonicalisation, admissibility filtering, or length selection. Consequently they do not bound the selected rule count or the peak number of rows in Todd–Coxeter. They are reported solely as reproducibility context for the calibration experiments.

E Conditional tools for the revised $\mathbf{PSL_2}$ parameters

Theorem 9 (Regular-action centralizer-orbit extraction). *Let G be a finite group of order N, given by a certified labelled right-regular action with d input-generator transition tables and identity row e. Let u be a supplied row, put*

$$C = C_G(u), \qquad c = |C|, \qquad m = [G : N_G(C)].$$

Then one can recover the conjugation action of G on the orbit of C, of degree m, using

$$O(dN + dmc \log c)$$

table lookups and

$$O(dN + mc)$$

words of storage. With hashing, the orbit phase takes expected $O(dmc)$ time apart from sorting the subgroup representatives. The algorithm uses no multiplication table and no oracle for arbitrary products.

Proof. Build a spanning tree of the right Cayley graph from e. For every input generator s, its left translation is determined by

$$L_s(e) = es, \qquad L_s(xs') = L_s(x)s'$$

along a tree edge labelled s'. Thus all L_s are constructed in $O(dN)$ lookups. Their inverse arrays give the left-generator transitions, from which we build a spanning tree of the left Cayley graph in the same bound.

Propagate L_u along the right tree from $L_u(e) = eu$. Propagate R_u along the left tree from $R_u(e) = eu$ using

$$R_u(L_s(x)) = L_s(R_u(x)).$$

Left and right translations commute, so both arrays are correct and cost $O(N)$ lookups. A scan then returns

$$C = \{x : L_u(x) = R_u(x)\}.$$

For every input generator s, the array

$$\gamma_s = L_s^{-1} R_s$$

is conjugation $x \mapsto s^{-1}xs$. Enumerate the orbit of the sorted c-tuple representing C under the γ_s. It contains exactly m subgroups by orbit–stabilizer. Each candidate requires c conjugation lookups and sorting, giving the stated time and storage bounds. Reading the image of every orbit member under each γ_s gives the induced generator permutations.

For public verification, label a row g by C^g, starting from $\lambda(e) = C$ and propagating

$$\lambda(gs) = \lambda(g)^s$$

along the right tree. Checking this identity on all generator edges, together with the m equal-size fibres, verifies the induced action against the regular action. No step materializes a map $(x, y) \mapsto xy$.

E.1 The July 2026 revision

The latest GRAFHEN paper explicitly acknowledges HEnbane and replaces its original S_{11} recommendation by $G = \mathrm{PSL}_2(343)$ in the projective action of degree 344 [8, Section 8]. It uses $E = \mathrm{PSL}_2(7)$ for binary encoding and a point stabiliser as the replacement for the original masking subgroup. The authors motivate the change by HEnbane's degree-dependent search and state a worst-case estimate of 2^{1720} for that attack model.

The relevant group order is

$$|\mathrm{PSL}_2(343)| = \frac{343(343^2 - 1)}{2} = 20{,}176{,}632,$$

roughly one half of $|S_{11}|$. The PSL implementation supplies the group-specific recognition and alignment path. The missing result is a target-size end-to-end evaluation on a two-side public corpus in the authors' KeyGen format.

Scope of the revised evaluation. The revised paper specifies the platform group, the encoding subgroup $E = \mathrm{PSL}_2(7)$, and the use of a degree-344 point stabilizer in place of the earlier masking subgroup. It does not provide a target-size public PSL rule corpus on which we can run an end-to-end evaluation. The results below therefore benchmark recognition and certification conditional on a regular table at the target group. Generated full-FP controls at $q \in \{7, 25, 27, 49, 169\}$ additionally exercise public-rule recovery and encryption-layer calibration; the $q = 169$ instance also exercises two-side alignment. The $k = 2$ admissibility-filtered corpus at $q = 169$ also reaches target order and is compared below. These experiments do not yet measure filtered-corpus convergence at $q = 343$. The benchmark following the revised recommendation remains an S_{11} instance, so it is not a performance measurement for those target-size stages.

E.2 Experimental methods and KeyGen boundary

The experimental implementation starts from generated regular actions and performs the unipotent–Borel blow-down, field coordinatization, pointwise matrix verification, and permutation-side order certification specified below. It covers

$$q \in \{7, 11, 13, 25, 27, 49, 81, 121, 125, 169, 343\};$$

the controls through $q = 169$ are GAP-generated, while the target instance is generated by a dedicated C program. The full measurements appear in Table 4. At the target,

$$|\mathrm{PSL}_2(343)| = \frac{343(343^2 - 1)}{2} = 20{,}176{,}632.$$

The regular-action generator takes 78 s and 924 MB; this cost is excluded from the subsequent recovery measurement.

q	$\lvert\mathrm{PSL}_2(q)\rvert$	rules	recover and recognize	calibrate and decrypt	total; peak RSS
7	168	251	0.08 s; 2.4 MB	0.13 s; 22.9 MB	0.23 s; 23 MB
25	7,800	12,331	0.98 s; 8.8 MB	1.19 s; 26.2 MB	2.20 s; 26 MB
27	9,828	15,579	1.28 s; 10.8 MB	1.54 s; 27.4 MB	2.86 s; 27 MB
49	58,800	93,683	10.09 s; 57.6 MB	10.47 s; 74.4 MB	20.68 s; 74 MB
169	2,413,320	3,747,158/side	79.1/80.2 s; 0.93/0.90 GiB	3.10 s; 78 MB	103 s; 0.93 GiB

Table 1. End-to-end recovery from generated unfiltered full-FP public corpora. Every run closes Todd–Coxeter at the target order, recognizes the projective tuple, and satisfies all four public-sandwich checks. The certification step itself takes at most 0.12 s and 2 MB through $q = 49$; at $q = 169$, checking all 3,747,158 relators takes 18.9 s and 13 MB. Calibration recovers the masking subgroup and the nonzero coset exactly, then decrypts the stated held-out ciphertexts with zero errors.

Unfiltered full-FP controls at $q = 7, 25, 27, 49$ exercise the one-side public-rule path. A two-side unfiltered full-FP control at $q = 169$ additionally exercises mixed-rule alignment; its measurements appear in Table 1. GRAFHEN KeyGen instead applies an admissibility filter to Froidure–Pin output, either after enumeration or during it. We therefore also measure a $k = 2$ admissibility-filtered corpus at $q = 169$, compared with the full-FP control in Table 2. To assess the pre-filter engineering boundary at the revised target, we implemented the corresponding Froidure–Pin construction in the degree-344 projective action on $\mathbb{P}^1(\mathbb{F}_{343})$, with $\mathbb{F}_{343} = \mathbb{F}_7[t]/(t^3 + 2)$. We sampled $d = 5$ generators as seeded random words of lengths 24–64 in an upper-root basis and Weyl generator (seed 20260723), using `libsemigroups_pybind11` 1.1.0. Table 3 gives the resulting Froidure–Pin sweep. All controls through $q = 289$ complete within the 24 GiB operating budget. The $q = 343$ run was terminated after its resident memory exceeded 32 GiB, before completion, filtering, or serialisation. The completed controls exhibit approximately linear memory growth in the enumerated group order. Thus the observed memory wall is a KeyGen phenomenon, distinct from the target-size post-enumeration measurement.

E.3 End-to-end public-rule recovery

The calibration implementation closes the masking subgroup directly in Python, so its measured memory and time are conservative. The recovered subgroup and coset agree exactly with the generation data in every control. This is a faithful instantiation of the point-stabilizer masking construction, rather than a claim about GRAFHEN's admissibility-filtered distribution. For the $q = 169$ row, the two independently recovered projective frames align through the mixed rules in 0.08 s at 14 MB; the resulting $\mathrm{P\Gamma L}_2$ element is unique.

E.4 KeyGen memory sweep

E.5 Recognition benchmark

Every benchmark accepts. The recognition time and memory track the linear-size regular-action input, while the certificate acts only on the degree-$q + 1$ projective

metric	full-FP	admissibility-filtered ($k = 2$)
public rules	3,747,158	534,609
generation time / file	24.8 s / 155 MB	16.1 s / 22 MB
selection	2.0 s	0.35 s
Todd–Coxeter time	78.3 s	85.1 s
peak rows	4,457,404	8,700,180
$M^*/\|G\|$	1.85	3.60
Todd–Coxeter peak RSS	0.93 GiB	0.75 GiB
recognition	0.83 s	0.89 s
all-relator sandwich check	18.9 s	2.8 s
held-out decryption	5,000/5,000	5,000/5,000

Table 2. Full-FP versus admissibility-filtered recovery at $\mathrm{PSL}_2(169)$. Both corpora close Todd–Coxeter at $|\mathrm{PSL}_2(169)| = 2{,}413{,}320$, yield certified projective tuples, and pass every relator in their respective public corpora. The filtered corpus has fewer resident rules and a faster full-corpus check, but it causes greater speculative-row overshoot.

q	$\|\mathrm{PSL}_2(q)\|$	enumerate	FP rules	rule scan	peak RSS
49	58,800	0.2 s	93,683	0.1 s	0.06 GiB
121	885,720	4.6 s	1,383,867	1.6 s	0.34 GiB
125	976,500	5.4 s	1,527,357	2.1 s	0.36 GiB
169	2,413,320	16.6 s	3,747,158	5.5 s	0.93 GiB
243	7,174,332	65.3 s	11,400,840	21.5 s	3.31 GiB
289	12,068,640	152.5 s	18,777,493	38.5 s	8.60 GiB
343	20,176,632	–	–	–	$>$ 32 GiB; killed

Table 3. Unfiltered Froidure–Pin enumeration sweep in the projective representation. The completed runs through $q = 289$ remain below the 24 GiB operating budget. The $q = 343$ process was terminated after exceeding 32 GiB, before the enumeration completed. This is a pre-filter resource measurement, not the size or cost of a published admissibility-filtered corpus.

action. In particular, at $q = 343$ the latter costs less than one second and 85 MB. These measurements establish that the representation-recovery components of Maverick are practical at the revised target once the regular action is available.

Remark 11 (Encryption-layer scope). The replacement of $E = S_{11}$ by $E = \mathrm{PSL}_2(7)$ and of the masking subgroup by a point stabilizer changes the concrete subgroup L and the published surjection $\pi : L \to E$. It does not change the calibration interface. When $\mathrm{Enc}(0) = 1_E$, the zero fibre is $\ker \pi$, and every labelled plaintext fibre is a coset of $\ker \pi$. Hence the common-coordinate tuple recovered by Maverick can be calibrated from public labelled encryptions exactly as in Proposition 2, provided these fibre conditions are checked on the revised corpus. The full-FP controls of Table 1 verify these conditions at $q = 7, 25, 27, 49, 169$; they remain unmeasured at $q = 343$.

Relation to the ciphertext-level discussion. Appendix C of the revised GRAFHEN paper responds to a separate IND-CPA distinguisher [7]. That distinguisher is distinct from representation recovery. Both HEnbane and Maverick nevertheless use the structural observation that, for key recovery, the semidirect-product layer adds no independent hardness once one per-side representation is recovered: the

q	$\lvert\mathrm{PSL}_2(q)\rvert$	degree	recognition and coordinatization	permutation-side certificate
7	168	8	0.02 s; 1.5 MB	0.01 s; 1.5 MB
11	660	12	0.03 s; 1.6 MB	0.01 s; 1.5 MB
13	1,092	14	0.04 s; 1.7 MB	0.01 s; 1.5 MB
25	7,800	26	0.02 s; 2.1 MB	0.03 s; 1.6 MB
27	9,828	28	0.01 s; 2.2 MB	0.01 s; 1.7 MB
49	58,800	50	0.04 s; 5.3 MB	0.01 s; 1.9 MB
81	265,680	82	0.09 s; 15.5 MB	0.02 s; 2.9 MB
121	885,720	122	0.46 s; 50.6 MB	0.02 s; 5.7 MB
125	976,500	126	0.46 s; 55.5 MB	0.03 s; 6.0 MB
169	2,413,320	170	1.48 s; 134 MB	0.05 s; 12.5 MB
343	20,176,632	344	35.0 s; 1.92 GiB	0.83 s; 85 MB

Table 4. Regular-action recognition benchmarks. Each run verifies the unipotent–Borel blow-down, field coordinatization, determinant-one matrix representatives, and pointwise generator action. The certificate column adds the BSGS and stabilizer-chain order checks. Controls through $q = 169$ are GAP-generated; $q = 343$ uses the dedicated C generator. The $q = 343$ regular-action generation cost (78 s and 924 MB) is excluded.

mixed relations determine the other side through conjugation. This does not assert that the semidirect product offers no protection against every ciphertext-only attack.

E.6 PSL recognition and alignment

Lemma 6 (Unipotent–Borel structure). *Let $q = p^f$ be odd and at least 5, and set $G = \mathrm{PSL}_2(q)$. If $u \in G$ has order p, then u is unipotent,*

$$|C_G(u)| = q, \qquad |N_G(C_G(u))| = \frac{q(q-1)}{2}.$$

Writing $U = C_G(u)$ and $B = N_G(U)$, the subgroup B is a Borel subgroup and the right-coset action on $B\backslash G$ has degree $q + 1$. It is equivalent to the standard projective action on $\mathbb{P}^1(\mathbb{F}_q)$.

Proof. The standard unipotent–Borel description of $\mathrm{PSL}_2(q)$ gives these facts [13]. An element of order p cannot be semisimple, since semisimple element orders divide $q - 1$ or $q + 1$, neither divisible by p. It is therefore conjugate to the image of

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

A direct matrix calculation in $\mathrm{SL}_2(q)$ shows that its centralizer, modulo the scalar centre, is the image of

$$\left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{F}_q \right\},$$

which has order q. Its normalizer is the image of the upper triangular matrices in $\mathrm{SL}_2(q)$, of order $q(q-1)/2$ in G. This is the stabilizer of ∞ in the projective action, so its index is $q + 1$ and the indicated coset action is the projective action.

Theorem 10 (Certified projective blow-down). *Let $q = p^f$ be odd and at least 5, set $G = \mathrm{PSL}_2(q)$ and $N = |G|$, and let a certified right-regular table for G have d distinguished input generators. Suppose the table supplies a nonidentity row u of order p. Then one can recover and certify the degree-$q+1$ projective action of G, together with the images of the input generators, using*

$$O(dN + dq^2)$$

generator-transition operations and

$$O(dN + q^2)$$

words of storage. Here "certified" has the target-order meaning of Remark 2: the source table is accompanied by the required relation and order certificate and has exactly N rows.

Proof. This is the specialization of Theorem 9 to a unipotent element of $\mathrm{PSL}_2(q)$; we spell out the resulting projective identification and bounds. Let e be the identity row and choose a rooted spanning tree of the right-regular Cayley graph. For each input generator s, construct its left translation L_s by setting $L_s(e) = s$ and propagating

$$L_s(xs) = L_s(x)s$$

along the tree. This is valid because left and right translations commute. It costs $O(dN)$ transitions and stores d arrays of length N. Invert these arrays and use them to construct a rooted spanning tree of the left Cayley graph, again in $O(dN)$ operations.

Construct L_u along the right tree from $L_u(e) = u$. Construct R_u along the left tree from $R_u(e) = u$, using

$$R_u(L_s(x)) = L_s(R_u(x)).$$

Both constructions are linear in N. Hence

$$U = \{x \in G : L_u(x) = R_u(x)\}$$

is computed by one scan of the rows. This equality holds precisely when $ux = xu$, so $U = C_G(u)$. The check $|U| = q$ certifies that U is a Sylow p-subgroup: q is the full p-part of $|G|$.

For every input generator s, the array

$$\gamma_s = L_s^{-1} R_s$$

is the right conjugation action $x \mapsto s^{-1}xs$. Enumerate the orbit of U under this right action, storing each subgroup as a sorted list of its q rows. By Lemma 6, the orbit consists of the $q+1$ Sylow-p subgroups and its stabilizer is the Borel subgroup $N_G(U)$. Applying every γ_s to these $q+1$ orbit points produces the required generator permutations. The orbit needs $O(q^2)$ words and $O(dq^2)$ transitions. Its size, together with $|U| = q$, is an exact certificate for the construction.

The action is faithful because its kernel is normal in the simple group G; equivalently, it is the right action on the cosets of the Borel subgroup. Lemma 6 therefore identifies it with the standard projective action. Summing the stated bounds proves the result.

The theorem separates deterministic recognition from discovery of u. A Las Vegas front end samples rows uniformly from the regular table, recovers the corresponding tree word, and accepts it when it is nonidentity and its pth power is the identity. Nontrivial order-p elements have density $\Theta(1/q)$ in $\mathrm{PSL}_2(q)$, so this needs $O(q)$ samples in expectation. The test is exact; the word length affects only this preliminary search, not the blow-down bound.

Lemma 7 (Blow-down certificate). *Under the hypotheses of Theorem 10, the projective action has an exact certificate of size $O(q^2 + dq)$ relative to the certified regular table. It consists of the order-p row u, a labelling*

$$\ell : \{x \in G : x \neq e,\ x \text{ has order } p\} \longrightarrow \{0, \dots, q\},$$

and the d induced permutations $\pi_s \in S_{q+1}$.

Proof. The prover labels the $q-1$ nonidentity elements of each Sylow-p subgroup by the corresponding orbit point. The verifier reconstructs the left translations from the regular table as in Theorem 10 and checks the following:

(i) the certificate lists $q^2 - 1$ distinct nonidentity rows, and every label in $\{0, \dots, q\}$ occurs exactly $q - 1$ times;

(ii) the fibre labelled 0, together with e, equals $C_G(u)$ and has order q;

(iii) for every labelled row x and every input generator s,

$$\ell(s^{-1}xs) = \pi_s(\ell(x));$$

(iv) the permutations π_s act transitively on $\{0, \dots, q\}$.

All checks use $O(dN + dq^2)$ generator transitions and $O(dN + q^2)$ words of storage.

Let $F_0 = \ell^{-1}(0) \cup \{e\}$. By (ii), $F_0 = C_G(u)$ is a Sylow-p subgroup. For a word w that sends 0 to a label b under the permutations π_s, (iii) gives

$$F_0^w \setminus \{e\} \subseteq \ell^{-1}(b).$$

Both sets have $q - 1$ elements by (i), so equality holds. By (iv), every fibre is therefore a conjugate of F_0. The PSL unipotent–Borel structure of Lemma 6 identifies these fibres with all Sylow-p subgroups and identifies (π_s) with the projective action.

This certificate verifies recognition relative to a certified regular action; it does not replace the separate task of publicly certifying that the selected presentation has the target order.

Lemma 8 (Stabilizer-chain certificate). *Let $H = \langle \sigma_1, \dots, \sigma_d \rangle \le \mathrm{Sym}(\Omega)$, and let $b_1, \dots, b_r$ be distinct points. For each i, suppose straight-line programs in the σ_j describe elements generating a subgroup*

$$K_i \le H_{b_1,\dots,b_{i-1}}$$

whose orbit of b_i has size n_i. Then

$$|H| \ge \prod_{i=1}^{r} n_i.$$

The straight-line programs and their orbit trees form a publicly checkable certificate of this lower bound.

Proof. The verifier evaluates every straight-line program, checks that its value fixes $b_1, \dots, b_{i-1}$, and computes the K_i-orbit of b_i. Hence

$$n_i \le [H_{b_1,\dots,b_{i-1}} : H_{b_1,\dots,b_i}]$$

by orbit–stabilizer. Multiplying these inequalities gives the claim.

Theorem 11 (Public sandwich certificate). *Let R be public relators on $a_1, \dots, a_d$, let $P_R = \langle a_1, \dots, a_d \mid R \rangle$, and set*

$$N = |\mathrm{PSL}_2(q)|.$$

Suppose the following publicly checkable artifacts are given.

(i) A replayable Todd–Coxeter trace over $\langle a_1 \rangle$, in the sense of Lemma 1, certifies $[P_R : \langle a_1 \rangle] \le n$, a derivation from R certifies $a_1^k = 1$, and $nk = N$.

(ii) A tuple $\sigma = (\sigma_1, \dots, \sigma_d) \in S_{q+1}^d$ satisfies every relator in R.

(iii) Straight-line programs in σ give a stabilizer-chain certificate, in the sense of Lemma 8, that $H = \langle \sigma \rangle$ has order at least N.

(iv) A field description of $\mathbb{F}_q$, a bijection $\lambda : \{1, \dots, q+1\} \to \mathbb{P}^1(\mathbb{F}_q)$, and matrices $M_i \in \mathrm{SL}_2(\mathbb{F}_q)$ satisfy

$$\lambda(\sigma_i(x)) = M_i \lambda(x) \qquad (1 \le i \le d,\ 1 \le x \le q+1).$$

Then the marked map $a_i \mapsto \sigma_i$ induces an isomorphism

$$P_R \cong H = \lambda^{-1} \mathrm{PSL}_2(q) \lambda.$$

Proof. Item (i) gives

$$|P_R| \le [P_R : \langle a_1 \rangle] \cdot |\langle a_1 \rangle| \le nk = N.$$

Item (ii) gives a surjection $P_R \twoheadrightarrow H$, while item (iii) gives $|H| \ge N$. Hence

$$N \le |H| \le |P_R| \le N,$$

so the marked surjection is an isomorphism. Item (iv) gives $H \le \lambda^{-1} \mathrm{PSL}_2(q) \lambda$, whose order is N; equality follows from $|H| = N$. This proves the claim.

For $\mathrm{PSL}_2(q)$, the lower bound in item (iii) can use the base $(\infty, 0, 1)$ and orbit sizes

$$q+1, \qquad q, \qquad (q-1)/2,$$

whose product is N. The trace is the only potentially linear-size artifact. If its producing run performs T table and relator-scan operations and has peak M live cosets, its replay uses $O(T+Md+dq)$ elementary operations and $O(Md+q\log q)$ words; its encoding has $\widetilde{O}(T)$ bits. The bound is deliberately parametric in M: an enumeration can overshoot the final index n, and an $O(nd)$ bound requires the additional property $M = O(n)$ (Remark 3). The stabiliser-chain and coordinate checks are lower order. The theorem separates two tasks that are often conflated: items (i)–(iii) link the selected public presentation to the recovered tuple, while item (iv) identifies that tuple with the standard projective action. A terminating rewriting system over $a_i^{\pm 1}$, with each rule derived from R, can replace item (i) when its number of irreducible words supplies the same upper bound; confluence is unnecessary for this purpose.

Theorem 12 (Intrinsic projective coordinatization). *Let $q = p^f \geq 5$ be odd, and let*

$$G = \langle \sigma_1, \dots, \sigma_d \rangle \leq S(\Omega), \qquad |\Omega| = q+1,$$

be permutation-isomorphic to the conjugation action of $\mathrm{PSL}_2(q)$ on its Sylow-p subgroups. Given only the generator permutations, one can deterministically construct either FAIL *or the following object:*

$$\big(m, \lambda, M_1, \dots, M_d\big),$$

where m is an irreducible degree-f polynomial over $\mathbb{F}_p$, $\lambda : \Omega \to \mathbb{P}^1(\mathbb{F}_p[X]/(m))$ is a bijection, and $M_i \in \mathrm{SL}_2(\mathbb{F}_p[X]/(m))$ satisfies

$$\lambda(\sigma_i(x)) = M_i \lambda(x) \qquad (x \in \Omega).$$

Under the stated hypothesis it does not fail, and

$$\lambda G \lambda^{-1} = \mathrm{PSL}_2\big(\mathbb{F}_p[X]/(m)\big).$$

The construction uses $O(dfq^2 + qf^3)$ permutation-entry and field operations and $O(q^2)$ words. The displayed object is an independently verifiable certificate: without any promise on G, checking irreducibility, bijection, determinant, and the $d(q+1)$ displayed equalities proves

$$\lambda G \lambda^{-1} \leq \mathrm{PSL}_2\big(\mathbb{F}_p[X]/(m)\big).$$

Proof. Fix $\omega_\infty \in \Omega$. A Schreier tree for the transitive action gives at most $d(q+1)$ Schreier generators for $B = G_{\omega_\infty}$ in $O(dq^2)$ entry operations and $O(q^2)$ space. The unipotent radical $U \lhd B$ is recovered intrinsically. Scan these generators for a nonidentity h. If h fixes only ω_∞, take $u_1 = h$. Otherwise it fixes ω_∞ and a second point ω_0. Choose a Schreier generator b moving ω_0 and put

$$u_1 = h(bhb^{-1})^{-1}.$$

The two factors have the same image in the abelian quotient B/U, so $u_1 \in U$. It is nontrivial because otherwise h and bhb^{-1} would have the same two fixed points. In the first case choose any $\omega_0 \neq \omega_\infty$.

Starting from u_1, close an $\mathbb{F}_p$-basis under conjugation by the Schreier generators of B. A span element is named by its image of ω_0: U acts regularly on $\Omega \setminus \{\omega_\infty\}$, so this image makes span membership a table lookup. The resulting nonzero B-invariant subspace is all of U. Indeed, in standard coordinates $B = U \rtimes T$, where T acts by the nonzero squares $S \subseteq \mathbb{F}_q^\times$, and

$$\mathbb{F}_p[S] = \mathbb{F}_q.$$

The latter equality follows because a proper subfield has at most $\sqrt{q}$ elements, whereas $S \cup \{0\}$ has $(q+1)/2 > \sqrt{q}$ elements. Thus U is an irreducible $\mathbb{F}_p B$-module. We obtain an $\mathbb{F}_p$-basis $u_1, \ldots, u_f$ of U, all q elements of U, and coordinates for the unique $u_x \in U$ satisfying $u_x(\omega_0) = x$.

Let $T = B_{\omega_0}$. Every Schreier generator s of B factors uniquely as $s = u_s t_s$, with $u_s(\omega_0) = s(\omega_0)$ and $t_s \in T$. The t_s generate T. Their conjugation actions on U give matrices $\alpha_{t_s} \in \mathrm{GL}_f(\mathbb{F}_p)$. Close these matrices under addition and multiplication to obtain the algebra

$$A = \mathbb{F}_p[\alpha_{t_s} : s].$$

In standard coordinates it is precisely $\mathbb{F}_p[S] = \mathbb{F}_q$. Check $\dim_{\mathbb{F}_p} A = f$, select an element $\gamma \in A$ with degree-f minimal polynomial m, and identify

$$F = \mathbb{F}_p[X]/(m) \simeq A, \qquad X \longmapsto \gamma.$$

Such an element exists because the union of the proper subfields has fewer than $f\sqrt{q}$ elements; a deterministic scan of A finds one in $O(qf^3)$ field operations.

The A-module U is one-dimensional. Using u_1 as a basis, map $c \in F$ to $\iota(c)(u_1) \in U$, and define

$$\lambda(\omega_\infty) = \infty, \qquad \lambda(x) = c \quad \Longleftrightarrow \quad \iota(c)(u_1) = u_x \quad (x \neq \omega_\infty).$$

For each σ_i, interpolate the unique projective matrix carrying $(\infty, 0, 1)$ to the images of the corresponding three labelled points. Its determinant has square class, so rescaling gives $M_i \in \mathrm{SL}_2(F)$. The final pointwise check either rejects or proves the displayed identities.

For completeness, identify the given action with the standard action and send ω_∞ to ∞ and ω_0 to 0. Then U becomes the translation group and T becomes multiplication by S. The preceding construction identifies A with $\mathbb{F}_q$ and makes λ the composition of a field isomorphism with a projective scaling. Both operations normalize the standard $\mathrm{PSL}_2(q)$, proving equality and ensuring that the checks pass. Conversely, the certificate checks themselves only assert that every conjugated generator belongs to $\mathrm{PSL}_2(F)$, which proves the promised unconditional soundness.

Remark 12. Theorem 12 supplies item (iv) of Theorem 11; combined with item (iii), containment becomes equality. It uses neither a group presentation nor words in the input generators. The certificate has $O(qf \log p)$ bits, and its verification uses $O(dq)$ field operations. In particular, this natural-action setting avoids the discrete-logarithm issue of black-box constructive recognition: the regular action of U supplies addition, while the torus reconstructs multiplication as an operator algebra.

For $q = 343$, Theorem 10 replaces the symmetric-group recognizer of Lemma 3. It returns a projective tuple up to a change of projective coordinates. Theorem 12 makes those coordinates explicit, hence supplies the standard projective copy needed for the alignment theorem below.

Lemma 9 (Projective centralizer and normalizer). *Let $q \geq 5$ be odd and let T be a degree-$q+1$ projective copy of $\mathrm{PSL}_2(q)$. Then*

$$C_{S_{q+1}}(T) = 1 \qquad \text{and} \qquad N_{S_{q+1}}(T) = \mathrm{P\Gamma L}_2(q)$$

in projective coordinates. Moreover, every subgroup of $\mathrm{P\Gamma L}_2(q)$ isomorphic to $\mathrm{PSL}_2(q)$ is its socle $\mathrm{PSL}_2(q)$.

Proof. The centralizer of a transitive permutation group embeds in the quotient of a point stabilizer's normalizer by that stabilizer. Here the point stabilizer is a self-normalizing Borel subgroup, so the centralizer is trivial. The normalizer statement is the standard description of the automorphism group of the projective action [4,13]. Finally, $\mathrm{PSL}_2(q)$ is perfect and

$$\mathrm{P\Gamma L}_2(q)/\mathrm{PSL}_2(q)$$

is solvable of order $2f$. An isomorphic copy of $\mathrm{PSL}_2(q)$ therefore maps trivially to this quotient and, by order, equals the socle.

Theorem 13 (Conditional projective-frame alignment). *Let $q = p^f \geq 5$ be odd. Let*

$$A = \langle \varphi_A(a_1), \dots, \varphi_A(a_d) \rangle \quad \text{and} \quad B = \langle \varphi_B(b_1), \dots, \varphi_B(b_{d'}) \rangle$$

be independently marked projective copies of $\mathrm{PSL}_2(q)$ on point sets Ω_A and Ω_B, each equipped with projective coordinates. For every i, j, let w_{ij} be a straight-line program of size at most L in the a-generators. Suppose that the substitutions

$$a_j \longmapsto w_{ij}(a)$$

induce automorphisms of A, and that the corresponding elements $\beta_i \in N_{S(\Omega_A)}(A)$ generate A. Suppose further that public unipotent words provide, on each side, fixed-point frames consisting of $0, 1, \infty$ and one point with coordinate generating $\mathbb{F}_q$ over $\mathbb{F}_p$.

Define an alignment to be a permutation $\gamma : \Omega_A \to \Omega_B$ satisfying the mixed equations

$$\gamma\varphi_A(w_{ij})\gamma^{-1} = \varphi_B(b_i)\big(\gamma\varphi_A(a_j)\gamma^{-1}\big)\varphi_B(b_i)^{-1} \qquad \text{for all } i, j. \tag{2}$$

There is at most one alignment. It can be found, or its nonexistence correctly reported, without enumerating the $2f|\mathrm{PSL}_2(q)|$ elements of $\mathrm{P\Gamma L}_2(q)$: after three frame points determine the projective-linear part, only the f field twists are tested. The computation uses

$$O(dq) \quad \text{preprocessing}, \qquad O\big(d'(qL+f)\big)$$

operations to recover the β_i, and $O(qL+f)$ operations to assemble γ. The output γ is verified by eq. (2) in $O(dd'qL)$ operations.

Proof. First consider soundness and uniqueness, for which the mixed equations are the definition of alignment: independently marked tuples do not supply a shared abstract element g to intertwine. If γ satisfies eq. (2), each $\varphi_B(b_i)$ conjugates every generator of

$$H = \gamma A \gamma^{-1}$$

into H. Finiteness gives equality, so $B \le N_{S(\Omega_B)}(H)$. By Lemma 9, B is the socle of this normalizer and hence $B = H$.

Put $\beta_i = \gamma^{-1}\varphi_B(b_i)\gamma$. The equations show that β_i implements the prescribed substitution on the generators of A. Such an element is unique: two choices differ by an element of $C_{S(\Omega_A)}(A)$, which is trivial by Lemma 9. Thus if γ_0 and γ_1 are alignments, then

$$\eta = \gamma_0^{-1}\gamma_1$$

centralizes every β_i. The β_i generate A, so $\eta \in C_{S(\Omega_A)}(A) = 1$. Therefore $\gamma_0 = \gamma_1$.

For construction, evaluate the substitutions on unipotent words. The unique fixed points of their images give the action of every β_i on the A-side frame. The images of $0, 1, \infty$ determine its $\mathrm{PGL}_2(q)$ part by sharp 3-transitivity. These three rational points are fixed by every field automorphism; evaluating at the fourth, primitive-coordinate point leaves exactly the f possible Galois twists and selects at most one of them. A missing or nonunique fourth-point match, or a non-distinct computed triple, is a rejection condition.

Applying the same construction to the B-side frame determines the relative coordinate change γ (equivalently, one may first construct γ^{-1}). Finally evaluate every mixed equation pointwise. This last pass both rejects inconsistent data and certifies the returned alignment. The stated costs follow from evaluating the straight-line programs on $q+1$ points and from the f twist tests. The certificate need only encode γ as a projective matrix class and a twist exponent; the verifier expands it and checks eq. (2), without using the auxiliary β_i or the fixed-point words.

Remark 13. Theorem 13 is a conditional group-theoretic replacement for the direct $\mathrm{P\Gamma L}_2(q)$ enumeration. The present PSL code implements and benchmarks the projective recognition, coordinatization, and permutation-side certification path through $q = 343$ (Table 4) and the complete one-side public-rule path through $q = 49$, together with a two-side aligned recovery at $q = 169$ (Table 1). It has not yet been exercised end to end at $q = 343$. In particular, the theorem specifies the mixed-relation alignment certificate; the current benchmark does not claim a target-size public-presentation evaluation of this construction.

Together with a certified regular table, these steps instantiate the centreless-group template of Theorem 2. The Borel-action recognizer and the projective-frame alignment theorem are the group-specific ingredients.

The remaining work is the target-size presentation side of the end-to-end experiment: establish target-order convergence from a $\mathrm{PSL}_2(343)$ corpus, then run mixed-rule alignment and encryption-layer calibration on independent revised instances. Section E.2 records the current KeyGen memory boundary.

F Background on the Todd–Coxeter algorithm

We recall the Todd–Coxeter coset-enumeration terminology (Todd–Coxeter [16]; Sims [15, Chapter 4]).

Coset table. A partial table has one row for each currently known coset and one column for each generator and inverse. A definition fills an undefined cell with a new row; a relator scan follows a word through defined cells; and a coincidence merges rows whose scans force them to agree. Implementations use union–find and deduction queues to propagate such merges.

Strategies. HLT-style enumeration prioritises definitions, whereas Felsch-style enumeration prioritises deductions discovered at nearly complete relator scans. The implementation alternates the two. Its use of the known target order is deliberately limited to acceptance: a table is accepted only after a complete closure pass, as stated in Theorem 6.

Cost accounting. For a fixed corpus, scan work is driven by the number of live rows, the number of relators, and their lengths. This motivates the output-sensitive accounting in Section 6.1. The peak live-row count depends substantially on presentation and strategy; no presentation-independent bound stated here predicts the behaviour of the filtered GRAFHEN corpora.